



Robot Project Overview



Introduction

The final robot project will be completed in pairs (or teams of three) using your Sparkfun electronics kits as well as some laser cut and 3D-printed components. Each of the tasks builds on the previous task-- the project culminates in each team creating a robot that can navigate a secret maze! Throughout the project, you will be keeping track of your engineering design process using a digital design notebook. This will go towards creating your project portfolio. Your portfolio will include flowcharts that visually explain how your robot works, AutoCAD drawings of components of your robot and your modular maze system, your lasercut robot pieces to enhance robot chassis and tell a story about the iterations you made and the process you followed along the way.



Objectives

- Build a functioning robot that can complete three increasingly complex challenges
- Apply algorithmic thinking through programming with Arduino/C++
- Design elements for the final robot challenge using the design process
- Take your AutoCAD skill to another level by designing and lasercutting components for your robot
- Communicate ideas graphically through flowcharts and AutoCAD drawings
- Communicate results professionally through the development of a portfolio
- Track the history of your design using a digital design notebook.



Design Tasks

- For the first challenge, you will assemble your robot and program it to move across the table in a straight line from one end to the other and stop.
- For the second challenge, you will update your program so that your robot can navigate a specified course that you will know ahead of time. This will include starting, stopping, turning left, and turning right.
- For the third challenge, you will update your program incorporating sensors so that the robot can navigate a specified course that you will know ahead of time (using sensors to make decisions as it drives!).
- For the fourth challenge (last day of class), you will refine your program so that the robot can navigate an unknown maze "autonomously" (using sensors to make decisions as it drives!).

- You will also be responsible for creating your own modular maze system that another team will use to test their robot.
- Laser cutting is a requirement for this project. You can complete it at any time to enhance your robot's maneuverability.

PROJECT BREAKDOWN

TASK	DESCRIPTION	POINTS	UNIT
Challenge 1: Move and stop	Build and program the robot to drive in a straight line, staying within a path, from end to end on the table and stop.	10	Team
Challenge 2: Navigate specified course	Build and program the robot to drive through the specified course using components from the Sparkfun kit. Avoid walls using sensors. It can be partially pre-programmed.	10	Team
Challenge 3: Navigate a known maze	Build and program the robot to drive autonomously in a known maze, avoiding walls using sensors.	25	Team
Modular maze	Design and build modular maze for another team using cardboard and other makerspace materials.	5	Team
Lasercut robot part	Please incorporate the best design from your individual laser cutting challenge into your team project. Remember, your individual designs will also contribute to your AutoCAD take-home challenge grade.	10	Team
Challenge 4: Navigate unknown maze	Build and program the robot to drive autonomously in an unknown maze, avoiding walls using sensors.	40	Team
Total		100	Team
Portfolio	Prepare a portfolio about your entire project describing your design process, iterations on your prototype, and your final robot.	50	Individual

**subject to small changes*

SUGGESTIONS FOR SUCCESS

- Find ways to be creative!
 - Make this YOUR project. What do you want to learn or get better at?? How can you do that??

- Try to get things to happen more than once-- look for accuracy AND repeatability.
 - Prototype and test A LOT.
 - Repeat after me: START EARLY! START EARLY!
- Plan to incorporate your design, drawings, coding and results into your project portfolio.
 - Aka TAKE LOTS OF PICTURES AND NOTES!
 - Sometimes the robot works well until it gets to class (we don't know why this happens).
- Record everything in your digital design notebook to keep track of things for your portfolio.
 - Create sections for the portfolio and fill them in as you are working on your robot.
 - Keep track of your design process and use tools from the class to create a problem statement, define your problem (using the techniques in the textbook such as a Duncker diagram or KTPA), and generate design alternatives (structured brainstorming techniques).
 - Draw some parts of the project in AutoCAD.
- Meet with your partner/team regularly and plan your time together (set goals, assign responsibility)
 - Work on both the design and the coding for the robot.
 - Prepare for the unexpected and setbacks.
 - Figure out what might take a long time (such as fabricating parts) and give yourself time to do it
 - Allow all team members to be involved in both coding and design, not just the "expert".
 - Focus on what you want to get better at, not what you're already good at.